Introduction to Machine Learning

Probability Theory for ML

Readings:
Mitchell Ch. 1, 2, 6.1 – 6.3
Murphy Ch. 2
Bishop Ch. 1 - 2

Slides are adopted from Matt Gormley, Rob Hall, Zahra Koochak and Jeremy Irvin

Outline

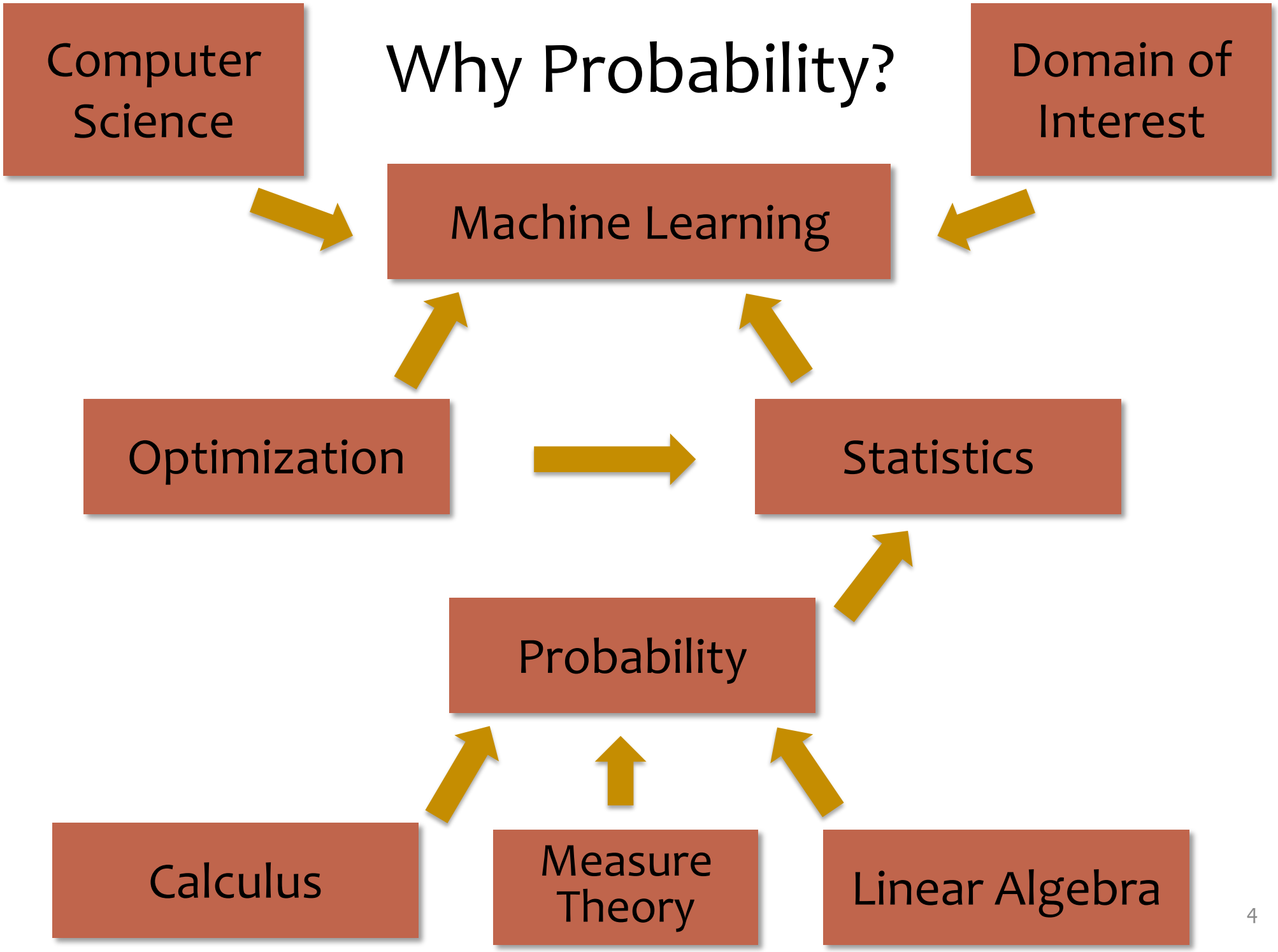
- **Motivation**
- **Probability Theory**
  - Sample space, Outcomes, Events
  - Kolmogorov's Axioms of Probability
- **Random Variables**
  - Random variables, Probability mass function (pmf), Probability density function (pdf), Cumulative distribution function (cdf)
  - Examples
  - Notation
  - Expectation and Variance
  - Joint, conditional, marginal probabilities
  - Independence
  - Bayes' Rule
- **Common Probability Distributions**
  - Beta, Dirichlet, etc.
- **Recap of Decision Trees**
  - Entropy
  - Information Gain
- **Probability in ML**

Why Probability?

The goal of this course is to provide you with a toolbox:



Why Probability?



PROBABILITY THEORY

Probability Theory: Definitions

Example

Example 1: Flipping a coin

Sample Space	Ω	{Heads, Tails}
Outcome	$\omega \in \Omega$	Example: Heads
Event	$E \subseteq \Omega$	Example: {Heads}
Probability	$P(E)$	$P(\{\text{Heads}\}) = 0.5$ $P(\{\text{Tails}\}) = 0.5$



Probability Theory: Definitions

Probability provides a science for inference about interesting events

Sample Space	Ω	The set of all possible outcomes
Outcome	$\omega \in \Omega$	Possible result of an experiment
Event	$E \subseteq \Omega$	Any subset of the sample space
Probability	$P(E)$	The non-negative number assigned to each event in the sample space

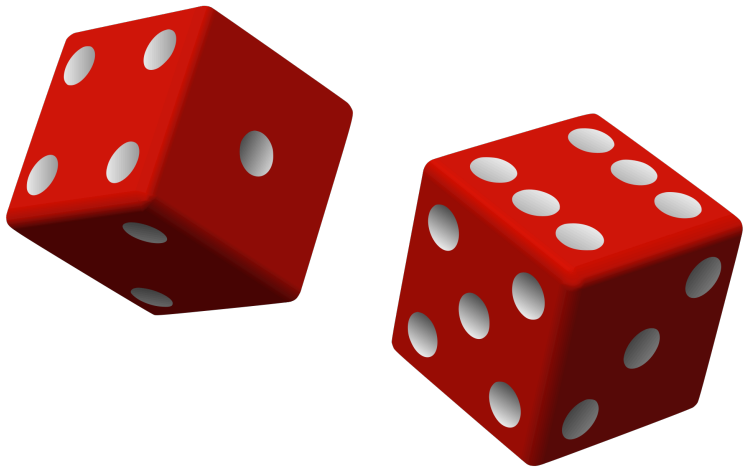
- Each outcome is unique
- Only one outcome can occur per experiment
- An outcome can be in multiple events
- An **elementary event** consists of exactly one outcome

Probability Theory: Definitions

Example

Example 2: Rolling a 6-sided die

Sample Space	Ω	{1,2,3,4,5,6}
Outcome	$\omega \in \Omega$	Example: 3
Event	$E \subseteq \Omega$	Example: {3} (the event "the die came up 3")
Probability	$P(E)$	$P(\{3\}) = 1/6$ $P(\{4\}) = 1/6$

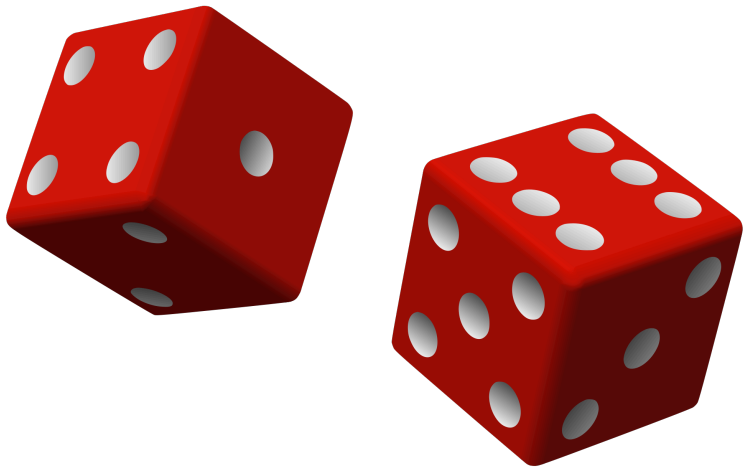


Probability Theory: Definitions

Example

Example 2: Rolling a 6-sided die

Sample Space	Ω	{1,2,3,4,5,6}
Outcome	$\omega \in \Omega$	Example: 3
Event	$E \subseteq \Omega$	Example: {2,4,6} (the event "the roll was even")
Probability	$P(E)$	$P(\{2,4,6\}) = 0.5$ $P(\{1,3,5\}) = 0.5$

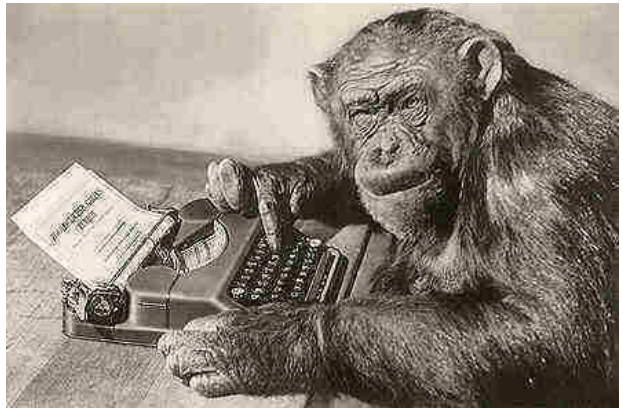


Probability Theory: Definitions

Example

Example 3: Timing how long it takes a monkey to reproduce Shakespeare

Sample Space	Ω	[0, +∞)
Outcome	$\omega \in \Omega$	Example: 1,433,600 hours
Event	$E \subseteq \Omega$	Example: [1, 6] hours
Probability	$P(E)$	$P([1,6]) = 0.000000000001$ $P([1{,}433{,}600, +\infty)) = 0.99$



Kolmogorov's Axioms

1. $P(E) \geq 0$, for all events E
2. $P(\Omega) = 1$
3. If $E_1, E_2, \ldots$ are disjoint, then
   $P(E_1 \text{ or } E_2 \text{ or } \ldots) = P(E_1) + P(E_2) + \ldots$

Kolmogorov's Axioms

1. $P(E) \geq 0$, for all events E
2. $P(\Omega) = 1$
3. If $E_1, E_2, \ldots$ are disjoint, then

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

All of probability can be derived from just these!

In words:

1. Each event has non-negative probability.
2. The probability that *some* event will occur is one.
3. The probability of the union of many disjoint sets is the sum of their probabilities

Deriving Probability Theorems

Axioms

1. $P(E) \geq 0$, for all events E
2. $P(\Omega) = 1$
3. If $E_1, E_2, \ldots$ are disjoint, then
   $P(E_1 \text{ or } E_2 \text{ or } \ldots) = P(E_1) + P(E_2) + \ldots$

Monotonicity: if A is a subset of B, then P(A) <= P(B)

Proof:

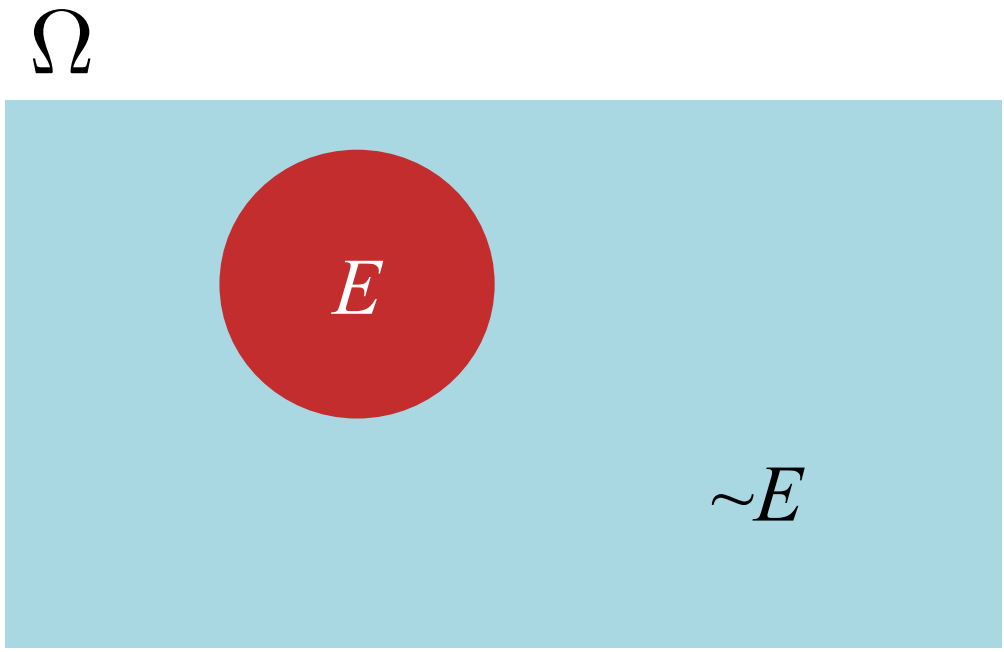
- A subset of B ➔ B = A + C for C=B-A
- A and C are disjoint ➔ P(B) = P(A or C)=P(A) + P(C)
- P(C) >= 0

- So P(B) >= P(A)

Slide adapted from William Cohen (10-601B, Spring 2016)

Probability Theory: Definitions

- The **complement** of an event E, denoted $\sim E$, is the event that E does not occur.



Deriving Probability Theorems

Axioms

1. $P(E) \geq 0$, for all events E
2. $P(\Omega) = 1$
3. If $E_1, E_2, \ldots$ are disjoint, then
   $P(E_1 \text{ or } E_2 \text{ or } \ldots) = P(E_1) + P(E_2) + \ldots$

Theorem: P(~A) = 1 - P(A)

Proof:

- P(A or ~A) = P(Ω) = 1
- A and ~A are disjoint ➔ P(A) + P(~A)=P(A or ~A)
  ➔ P(A) + P(~A) = 1

  ….then solve for P(~A)

Slide adapted from William Cohen (10-601B, Spring 2016)

Deriving Probability Theorems

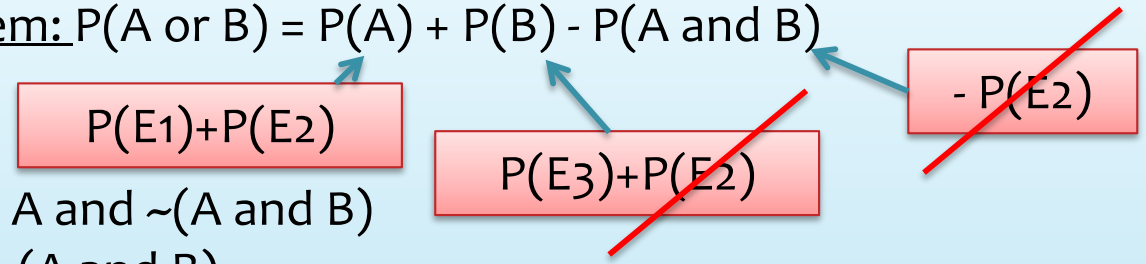
Axioms

1. $P(E) \geq 0$, for all events E
2. $P(\Omega) = 1$
3. If $E_1, E_2, \ldots$ are disjoint, then
   $P(E_1 \text{ or } E_2 \text{ or } \ldots) = P(E_1) + P(E_2) + \ldots$

Theorem: P(A or B) = P(A) + P(B) - P(A and B)

Proof:

- E1 = A and ~(A and B)
- E2 = (A and B)
- E3 = B and ~(A and B)
- E1 or E2 or E3 = A or B and E1, E2, E3 disjoint ➔
  P(A or B) = P(E1) + P(E2) + P(E3)
- further P(A) = P(E1) + P(E2) and P(B) = P(E3) + P(E2)
- ...



Slide adapted from William Cohen (10-601B, Spring 2016)

These Axioms are Not to be Trifled With

- Andrew Moore

- There have been many many other approaches to understanding "uncertainty":
    - Fuzzy Logic, three-valued logic, Dempster-Shafer, non-monotonic reasoning, …
- 40 years ago people in AI argued about these; now they mostly don't
  - *Any scheme* for combining uncertain information, uncertain "beliefs", etc,… really should obey these axioms to be internally consistent (from Jayne, 1958; Cox 1930's)
  - If you gamble based on "uncertain beliefs", then [you can be exploited by an opponent] ⇔ [your uncertainty formalism violates the axioms] - di Finetti 1931 (the "Dutch book argument")



RANDOM VARIABLES

Random Variables: Definitions

Random Variable	X (capital letters)	Def 1: Variable whose possible values are the outcomes of a random experiment
Value of a Random Variable	x (lowercase letters)	The value taken by a random variable

Random Variables: Definitions

Random Variable	X	Def 1: Variable whose possible values are the outcomes of a random experiment
Discrete Random Variable	X	Random variable whose values come from a countable set (e.g. the natural numbers or {True, False})
Continuous Random Variable	X	Random variable whose values come from an interval or collection of intervals (e.g. the real numbers or the range (3, 5))

Random Variables: Definitions

Random Variable	X	Def 1: Variable whose possible values are the outcomes of a random experiment Def 2: A measureable function from the sample space to the real numbers: $X : \Omega \to E$
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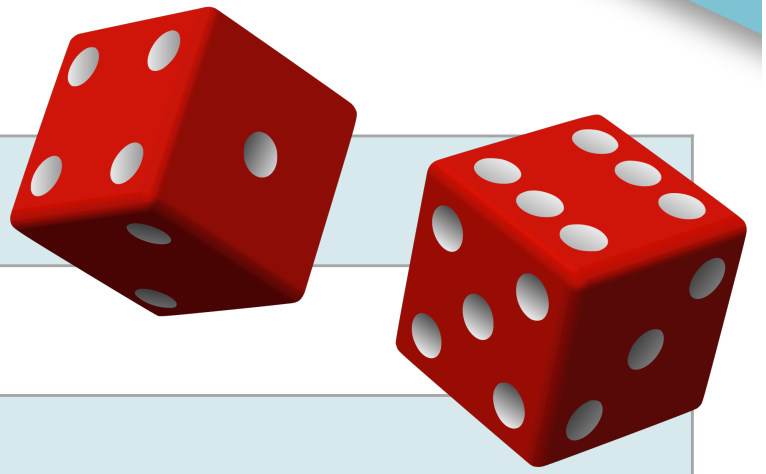
Random Variables: Definitions

Discrete Random Variable	X	Random variable whose values come from a countable set (e.g. the natural numbers or {True, False})
Probability mass function (pmf)	$p(x)$	Function giving the probability that discrete r.v. X takes value x. $$p(x) := P(X = x)$$

Random Variables: Definitions

Example

Example 2: Rolling a 6-sided die

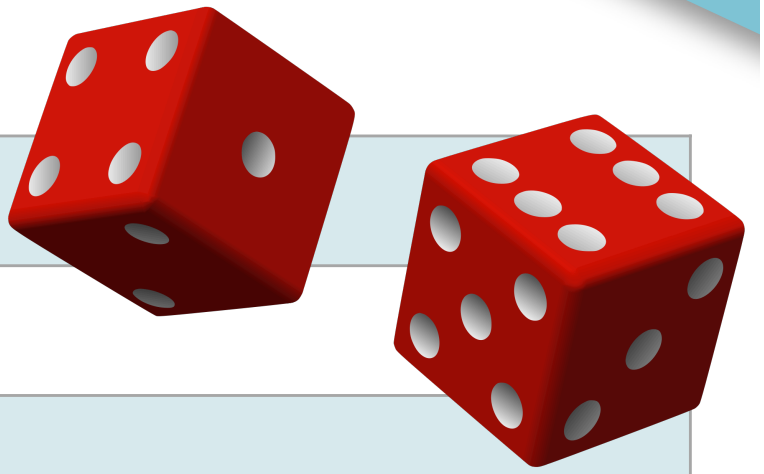


Sample Space	Ω	{1,2,3,4,5,6}
Outcome	$\omega \in \Omega$	Example: 3
Event	$E \subseteq \Omega$	Example: {3} (the event "the die came up 3")
Probability	$P(E)$	$P(\{3\}) = 1/6$ $P(\{4\}) = 1/6$

Random Variables: Definitions

Example

Example 2: Rolling a 6-sided die

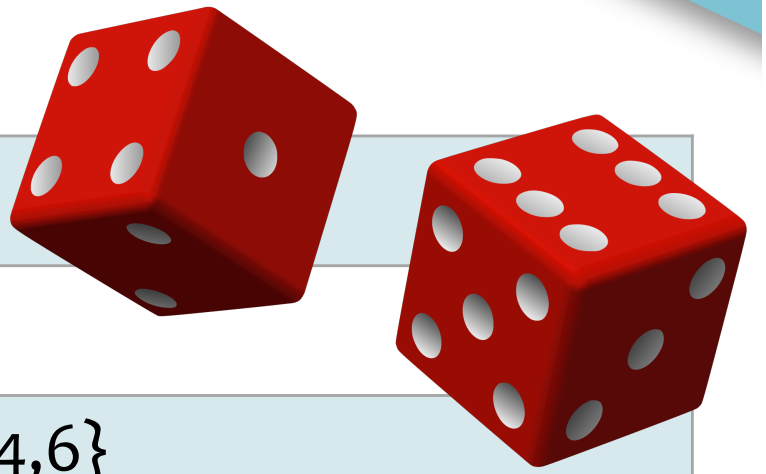


Sample Space	Ω	{1,2,3,4,5,6}
Outcome	$\omega \in \Omega$	Example: 3
Event	$E \subseteq \Omega$	Example: {3} (the event "the die came up 3")
Probability	$P(E)$	$P(\{3\}) = 1/6$ $P(\{4\}) = 1/6$
Discrete Random Variable	X	Example: The value on the top face of the die.
Prob. Mass Function (pmf)	$p(x)$	p(3) = 1/6 p(4) = 1/6

Random Variables: Definitions

Example

Example 2: Rolling a 6-sided die



Sample Space	Ω	{1,2,3,4,5,6}
Outcome	$\omega \in \Omega$	Example: 3
Event	$E \subseteq \Omega$	Example: {2,4,6} (the event "the roll was even")
Probability	$P(E)$	$P(\{2,4,6\}) = 0.5$ $P(\{1,3,5\}) = 0.5$
Discrete Random Variable	X	Example: 1 if the die landed on an even number and 0 otherwise
Prob. Mass Function (pmf)	$p(x)$	p(1) = 0.5 p(0) = 0.5

Random Variables: Definitions

Discrete Random Variable	X	Random variable whose values come from a countable set (e.g. the natural numbers or {True, False})
Probability mass function (pmf)	$p(x)$	Function giving the probability that discrete r.v. X takes value x. $p(x) := P(X = x)$

Random Variables: Definitions

Continuous Random Variable	X	Random variable whose values come from an interval or collection of intervals (e.g. the real numbers or the range (3, 5))
Probability density function (pdf)	$f(x)$	Function returns a nonnegative real indicating the relative likelihood that a continuous r.v. X takes value x

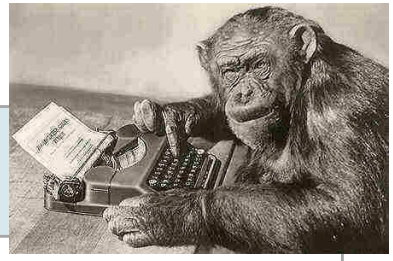
- For any continuous random variable: $P(X = x) = 0$
- Non-zero probabilities are only available to intervals:

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

Random Variables: Definitions

Example

Example 3: Timing how long it takes a monkey to reproduce Shakespeare

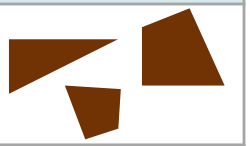


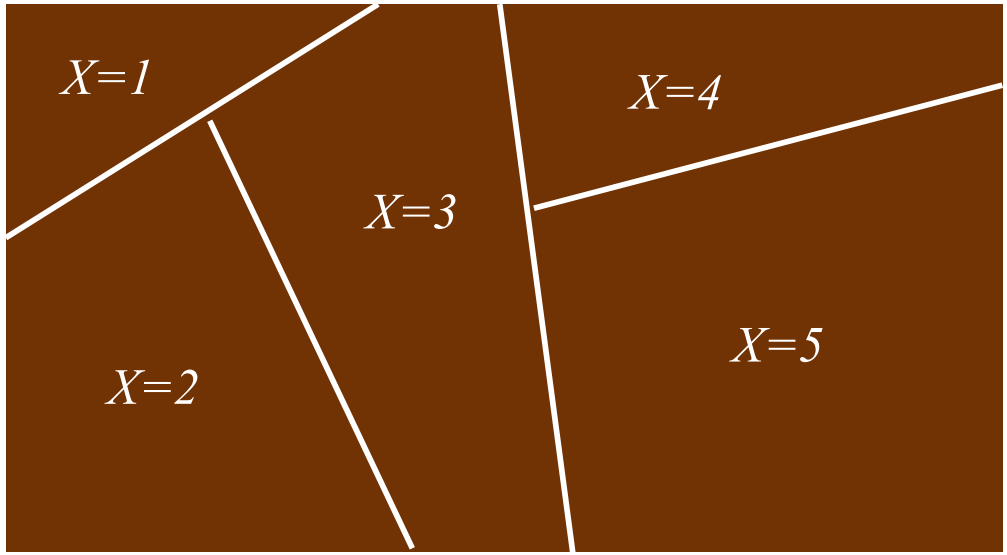
Sample Space	Ω	[0, +∞)
Outcome	$\omega \in \Omega$	Example: 1,433,600 hours
Event	$E \subseteq \Omega$	Example: [1, 6] hours
Probability	$P(E)$	$P([1,6]) = 0.000000000001$ $P([1{,}433{,}600, +\infty)) = 0.99$
Continuous Random Var.	X	Example: Represents time to reproduce (not an interval!)
Prob. Density Function	$f(x)$	Example: Gamma distribution

Random Variables: Definitions

Example

"Region"-valued Random Variables

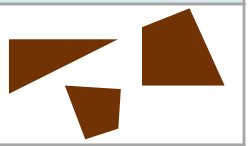
Sample Space	Ω	{1,2,3,4,5}
Events	x	The sub-regions 1, 2, 3, 4, or 5 
Discrete Random Variable	X	Represents a random selection of a sub-region
Prob. Mass Fn.	$P(X=x)$	Proportional to size of sub-region



Random Variables: Definitions

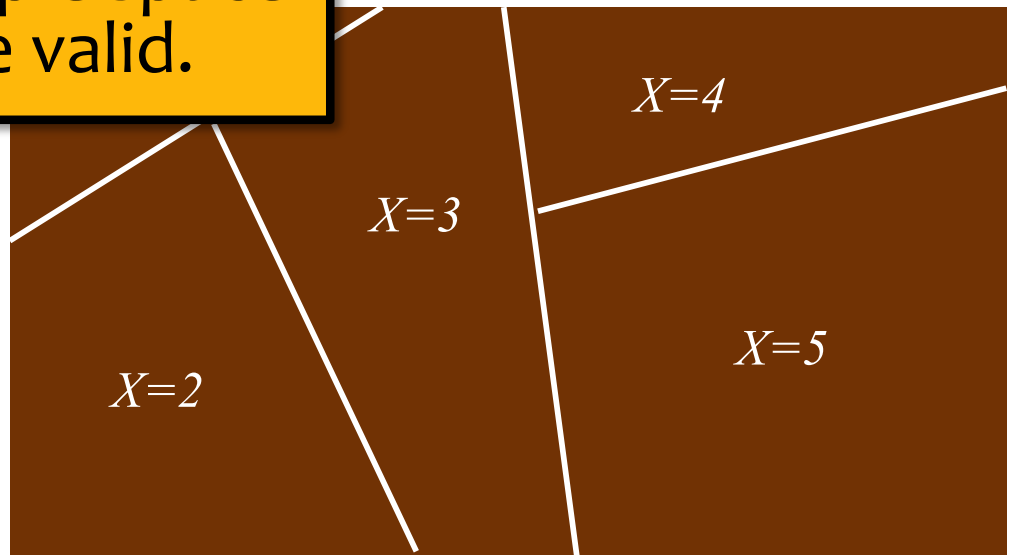
Example

"Region"-valued Random Variables

Sample Space	Ω	All points in the region: 
Events	x	The sub-regions 1, 2, 3, 4, or 5 
D **d**		R esents a random selection of a sub-region
P		Proportional to size of sub-region

Recall that an event is any subset of the sample space.

So both definitions of the sample space here are valid.



Random Variables: Definitions

Cumulative distribution function	$F(x)$	Function that returns the probability that a random variable X is less than or equal to x: $$F(x) = P(X \leq x)$$

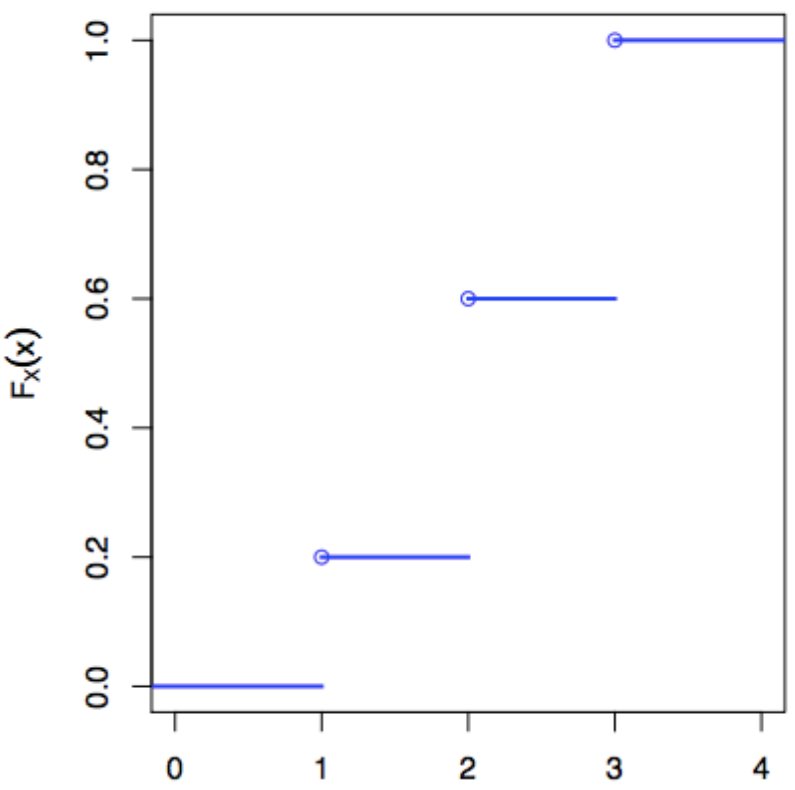
- For **discrete** random variables:

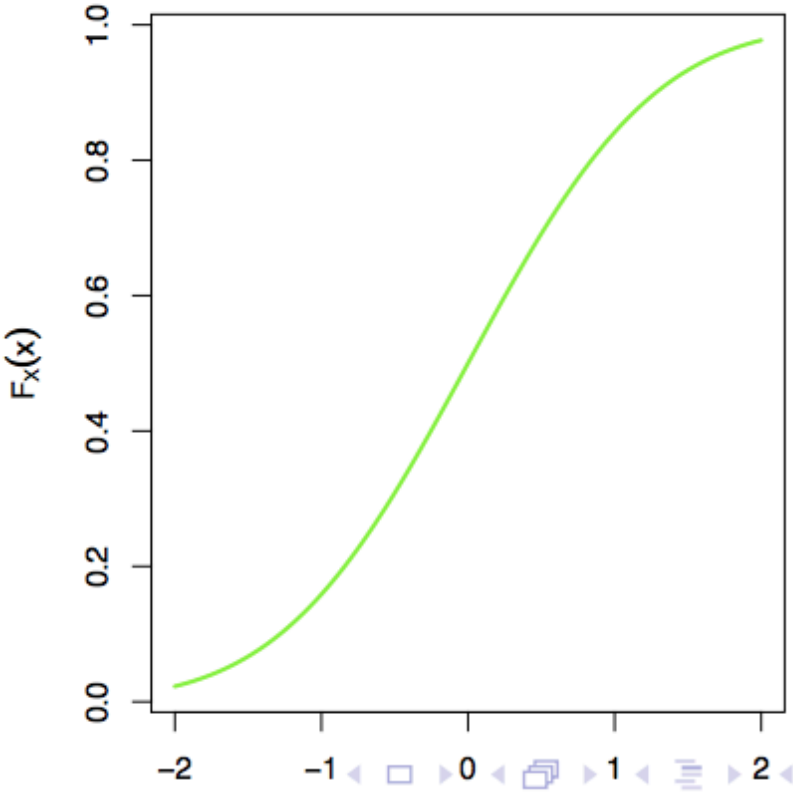
$$F(x) = P(X \leq x) = \sum_{x' < x} P(X = x') = \sum_{x' < x} p(x')$$

- For **continuous** random variables:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(x')dx'$$

- Then, the probability measure associated with the RV is completely characterized by its **cumulative distribution function (CDF)**: $F_X(x) = P(X \leq x)$.
- If two RVs have the same CDF we call then **identically distributed**.
- We say $X \sim F_X$ or $X \sim f_X$ (f_X coming soon) to indicate that X has the distribution specified by F_X (resp, f_X).





Continuous Distributions

- When the CDF is continuous we may consider its derivative $f_x(x) = \frac{d}{dx} F_X(x)$.
- This is called the **probability density function (PDF)**.
- The probability of an interval (a, b) is given by $P(a < X < b) = \int_a^b f_X(x)\, dx$.
- The probability of any specific point c is zero: $P(X = c) = 0$ (why?).

Random Variables and Events

Question: Something seems wrong…

- We defined $P(E)$ (the capital 'P') as a function mapping *events* to probabilities
- So why do we write $P(X=x)$?
- A good guess: $X=x$ is an event…

Random Variable	Def 2: A measureable function from the sample space to the real numbers: $X : \Omega \to \mathbb{R}$

Answer: $P(X=x)$ is just shorthand!

These sets are events!

Example 1:

$$P(X = x) \equiv P(\{\omega \in \Omega : X(\omega) = x\})$$

Example 2:

$$P(X \leq 7) \equiv P(\{\omega \in \Omega : X(\omega) \leq 7\})$$

Notational Shortcuts

A convenient shorthand:

$$P(A|B) = \frac{P(A,B)}{P(B)}$$

$\Rightarrow$ For all values of a and b:

$$P(A = a|B = b) = \frac{P(A = a, B = b)}{P(B = b)}$$

Expectation and Variance

The **expected value** of X is $E[X]$. Also called the mean.

- Discrete random variables:

  Suppose X can take any value in the set $\mathcal{X}$.

  $$E[X] = \sum_{x \in \mathcal{X}} x p(x)$$

- Continuous random variables:

  $$E[X] = \int_{-\infty}^{+\infty} x f(x) dx$$

Expectation and Variance

The **variance** of X is $Var(X)$.

$$Var(X) = E[(X - E[X])^2]$$

$$\mu = E[X]$$

- Discrete random variables:

$$Var(X) = \sum_{x \in \mathcal{X}} (x - \mu)^2 p(x)$$

- Continuous random variables:

$$Var(X) = \int_{-\infty}^{+\infty} (x - \mu)^2 f(x) dx$$

Variance

We may consider the **variance** of a distribution:

$$\text{Var}(X) = E(X - EX)^2$$

This may give an idea of how "spread out" a distribution is.

A useful alternate form is:

$$\begin{aligned} E(X - EX)^2 &= E[X^2 - 2XE(X) + (EX)^2] \\ &= E(X^2) - 2E(X)E(X) + (EX)^2 \\ &= E(X^2) - (EX)^2 \end{aligned}$$

Variance

Variance is non-linear but the following holds:

$$\mathrm{Var}(aX) = E(aX - E(aX))^2 = E(aX - aEX)^2 = a^2 E(X - EX)^2 = a^2 \mathrm{Var}(X)$$

$$\mathrm{Var}(X+c) = E(X+c-E(X+c))^2 = E(X-EX+c-c)^2 = E(X-EX)^2 = \mathrm{Var}(X)$$

$$\begin{aligned} \mathrm{Var}(X+Y) &= E(X - EX + Y - EY)^2 \\ &= \underbrace{E(X-EX)^2}_{\mathrm{Var}(X)} + \underbrace{E(Y-EY)^2}_{\mathrm{Var}(Y)} + 2\underbrace{E(X-EX)(Y-EY)}_{\mathrm{Cov}(X,Y)} \end{aligned}$$

So when X, Y are independent we have:

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

(why?)

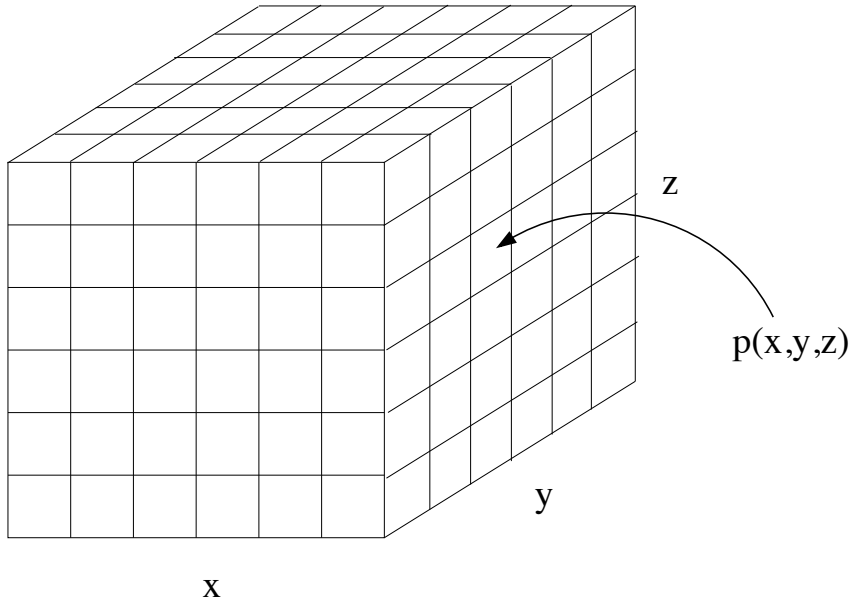
Multiple Random Variables

- Joint probability
- Marginal probability
- Conditional probability

Joint Probability

- Key concept: two or more random variables may interact. Thus, the probability of one taking on a certain value depends on which value(s) the others are taking.
- We call this a joint ensemble and write

$$p(x, y) = \text{prob}(X = x \text{ and } Y = y)$$



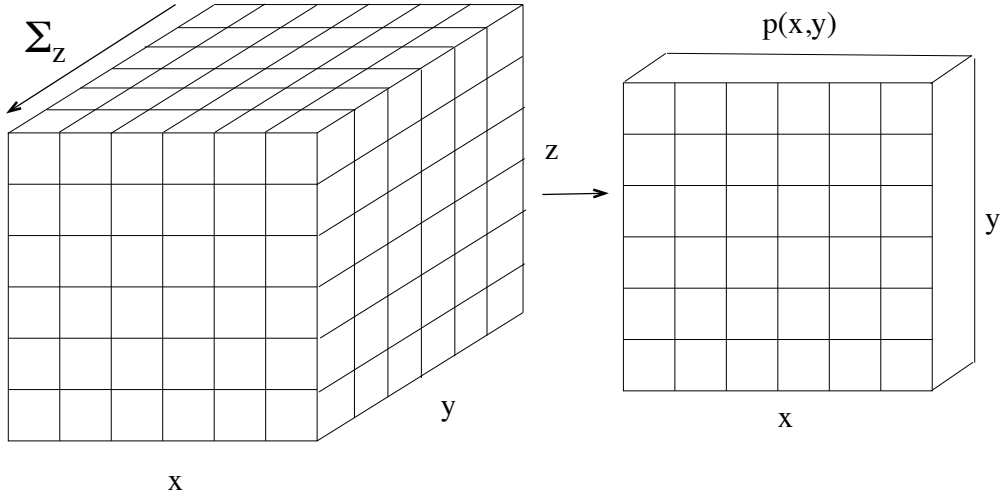
Slide from Sam Roweis (MLSS, 2005)

Marginal Probabilities

- We can "sum out" part of a joint distribution to get the *marginal distribution* of a subset of variables:

$$p(x) = \sum_y p(x, y)$$

- This is like adding slices of the table together.



- Another equivalent definition: $p(x) = \sum_y p(x|y)p(y)$.

Slide from Sam Roweis (MLSS, 2005)

Two Random Variables

Bivariate CDF

$$F_{XY}(x, y) = P(X \leq x, Y \leq y)$$

Bivariate PMF

$$p_{XY}(x, y) = P(X = x, Y = y)$$

Marginal PMF

$$p_X(x) = \sum_y p_{XY}(x, y)$$

Bivariate PDF

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x,y)}{\partial x \partial y}$$

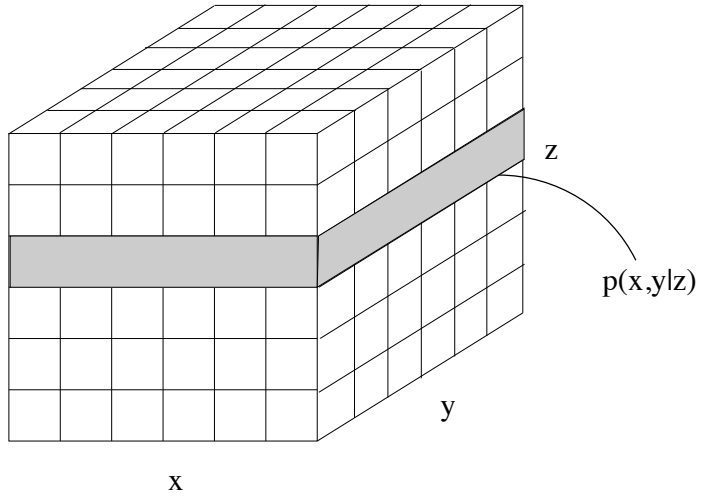
Marginal PDF

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

Conditional Probability

- If we know that some event has occurred, it changes our belief about the probability of other events.
- This is like taking a "slice" through the joint table.

$$\boxed{p(x|y) = p(x,y)/p(y)}$$

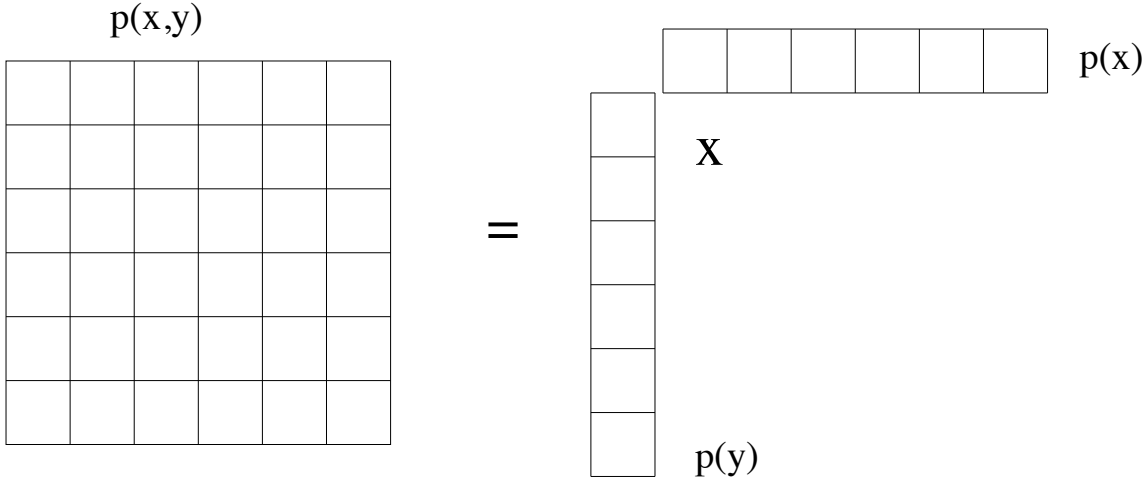


Slide from Sam Roweis (MLSS, 2005)

Independence and Conditional Independence

- Two variables are independent iff their joint factors:

$$p(x, y) = p(x)p(y)$$



- Two variables are conditionally independent given a third one if for all values of the conditioning variable, the resulting slice factors:

$$p(x, y|z) = p(x|z)p(y|z) \qquad \forall z$$

Slide from Sam Roweis (MLSS, 2005)

Example for independent random variables

- Spin a spinner numbered 1 to 7, and toss a coin. What is the probability of getting an odd. number on the spinner and a tail on the coin?

$$p_{XY}(x, y) = p_X(x)p_Y(y) = \frac{1}{2} \times \frac{4}{7} = \frac{2}{7}$$



Covariance of two random variables X and Y

$$Cov[x, y] = E[(x - E[x])(y - (E[y]))]$$
$$= E(XY) - E(X)E(Y)$$

If X and Y are independent, then:

$$E(XY) = E(X)E(Y) \rightarrow Cov[x, y] = 0$$

$$Var[X + Y] = [E(X + Y)]^2 - E((X + Y)^2)$$
$$Var[X + Y] = Var[X] + Var[Y] + 2Cov[X, Y]$$

Definition of Conditional Probability

$$P(A \mid B) = \frac{P(A \wedge B)}{P(B)}$$

Corollary: The Chain Rule

$$P(A \wedge B) = P(A \mid B)\, P(B)$$

Slide from William Cohen (10-601B, Spring 2016)

BAYES' RULE

posterior

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

prior

Bayes' rule

$$P(B|A) = \frac{P(A|B) * P(B)}{P(A)}$$



Bayes, Thomas (1763) An essay towards solving a problem in the doctrine of chances. *Philosophical Transactions of the Royal Society of London,* **53:370-418**

… by no means merely a curious speculation in the doctrine of chances, but necessary to be solved in order to a sure foundation for all our reasonings concerning past facts, and what is likely to be hereafter…. necessary to be considered by any that would give a clear account of the strength of *analogical* or *inductive reasoning*…

Slide from William Cohen (10-601B, Spring 2016)

Bayes' Theorem

- Given the conditional probability of an event $P(x|y)$
- Want to find the "reverse" conditional probability, $P(y|x)$

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

where: $P(x) = \sum_{y' \in value\ y} P(x|y')P(y')$

X and Y are continuous

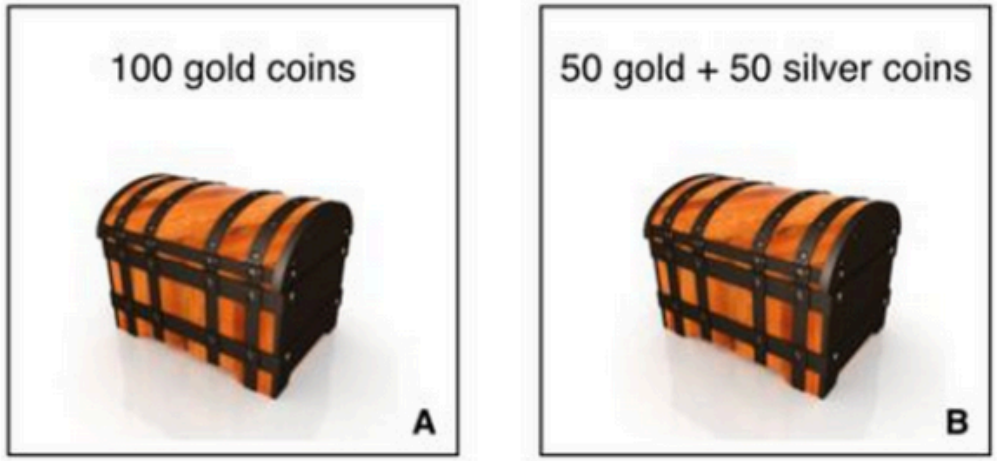
$$f(y|x) = \frac{f(x|y)f(y)}{f(x)}$$

where: $f(x) = \int_{y' \in value\ y} f(x|y')f(y')dy'$

Example for Bayes Rule

- You randomly choose a treasure chest to open, and then randomly choose a coin from that treasure chest. If the coin you choose is gold, then what is the probability that you choose chest A?

$$a)\frac{1}{3} \qquad b)\frac{2}{3} \qquad c)1 \qquad d)None$$



COMMON PROBABILITY DISTRIBUTIONS

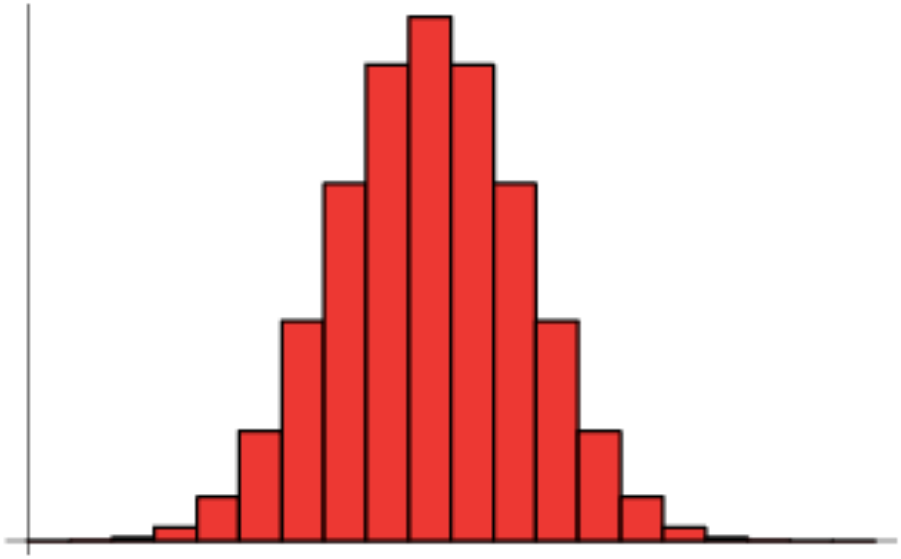
Common Probability Distributions

- For Discrete Random Variables:
  - Bernoulli
  - Binomial
  - Multinomial
  - Categorical
  - Poisson
- For Continuous Random Variables:
  - Exponential
  - Gamma
  - Beta
  - Dirichlet
  - Laplace
  - Gaussian (1D)
  - Multivariate Gaussian

Bernoulli Distribution

- e.g., general discrete PMF: $f_X(x_i) = \theta_i,\ \sum_i \theta_i = 1, \theta_i \geq 0.$
- e.g., bernoulli distribution: $X \in \{0, 1\},\ f_X(x) = \theta^x(1-\theta)^{1-x}$
- A general model of binary outcomes (coin flips etc.).

Binomial Distribution



The binomial distribution gives the discrete probability distribution $P_p(n \mid N)$ of obtaining exactly n successes out of N Bernoulli trials (where the result of each Bernoulli trial is true with probability p and false with probability $q = 1 - p$). The binomial distribution is therefore given by

$$P_p(n \mid N) = \binom{N}{n} p^n q^{N-n}$$
$$= \frac{N!}{n!\,(N-n)!} p^n (1-p)^{N-n},$$

where $\binom{N}{n}$ is a binomial coefficient. The above plot shows the distribution of n successes out of $N = 20$ trials with $p = q = 1/2$.

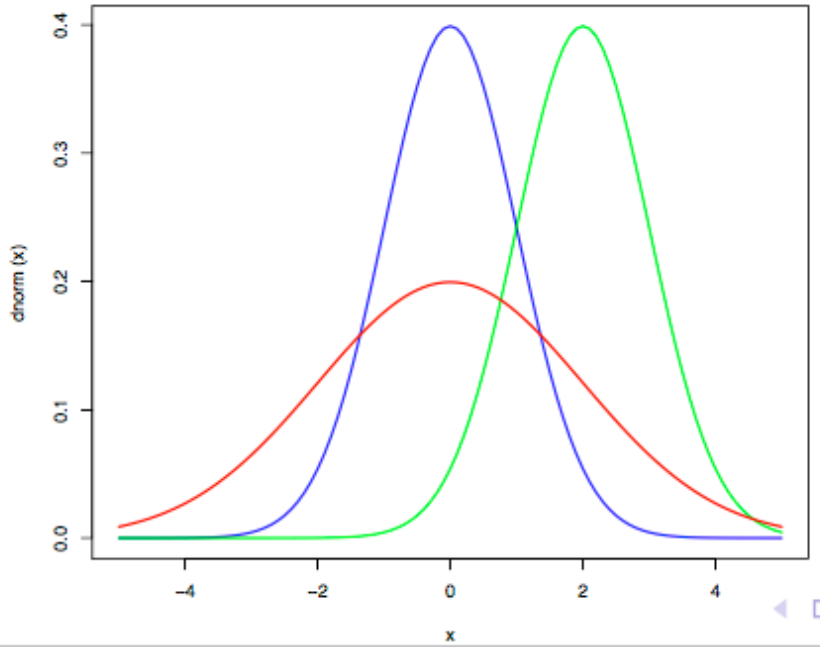
Slide from http://mathworld.wolfram.com/BinomialDistribution.html

Multinomial Distribution

- e.g., multinomial distribution:
  $X \in \mathbb{N}^d$, $\sum_{i=1}^{d} x_i = n$, $f_X(x) = \frac{n!}{x_1!x_2!\cdots x_d!} \prod_{i=1}^{d} \theta_i^{x_i}$.
- Sometimes used in text processing (dimensions correspond to words, n is the length of a document).

Continuous Distributions

- e.g., Uniform distribution: $f_X(x) = \frac{1}{b-a} \cdot 1_{(a,b)}(x)$
- e.g., Gaussian aka "normal:" $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\{\frac{(x-\mu)^2}{2\sigma^2}\}$
- Note that both families give probabilities for every interval on the real line, yet are specified by only two numbers.



Example Distributions

Distribution	PDF or PMF	Mean	Variance
$Bernoulli(p)$	$\begin{cases} p, & \text{if } x = 1 \\ 1-p, & \text{if } x = 0. \end{cases}$	p	$p(1-p)$
$Binomial(n,p)$	$\binom{n}{k} p^k (1-p)^{n-k}$ for $k = 0, 1, ..., n$	np	$np(1-p)$
$Geometric(p)$	$p(1-p)^{k-1}$ for $k = 1, 2, ...$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
$Poisson(\lambda)$	$\frac{e^{-\lambda}\lambda^k}{k!}$ for $k = 0, 1, ...$	λ	λ
$Uniform(a,b)$	$\frac{1}{b-a}$ for all $x \in (a,b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
$Gaussian(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ for all $x \in (-\infty, \infty)$	μ	σ^2
$Exponential(\lambda)$	$\lambda e^{-\lambda x}$ for all $x \geq 0, \lambda \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

Multivariant Gaussian (Normal) distribution

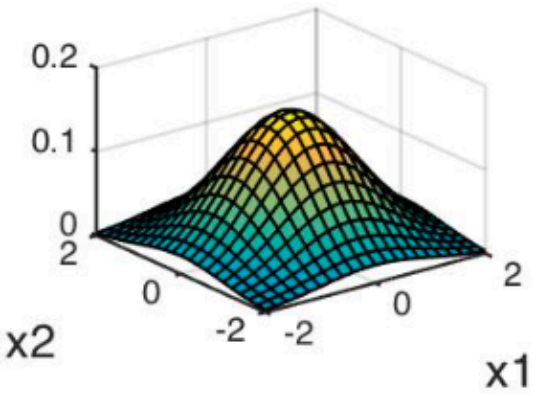
$x \in \mathbb{R}^n$. Model $p(x_1), p(x_2), ...etc.$ at the same time. Parameters $:\mu \in \mathbb{R}^n, \Sigma \in \mathbb{R}^{n \times n}(\text{covariancematrix})$

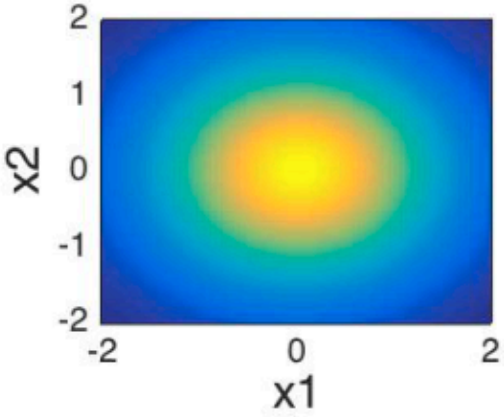
$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{\frac{1}{2}}} exp(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu))$$

Multivariant Gaussian (Normal examples)

$$\Sigma = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}$$

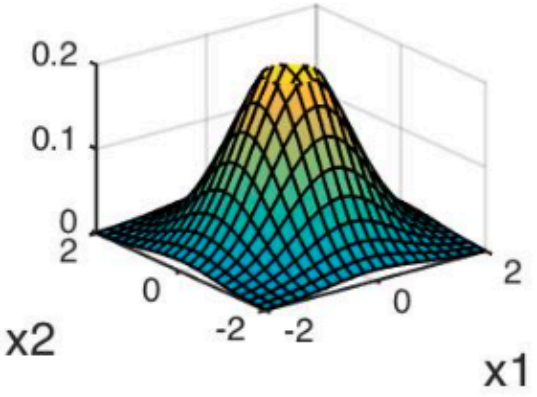
$$\mu = [0 \;\; 0]^T$$

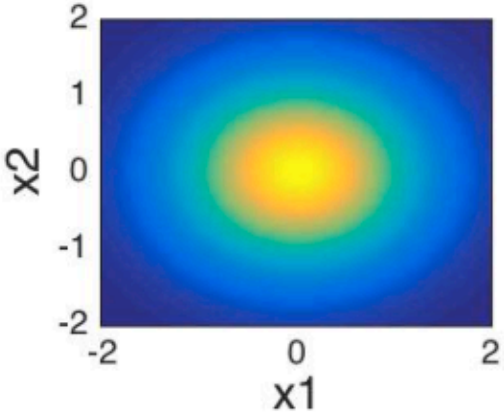




$$\Sigma = \begin{matrix} 0.7 & 0 \\ 0 & 0.7 \end{matrix}$$

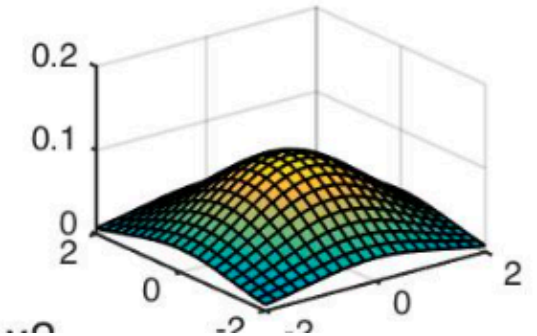
$$\mu = [0 \;\; 0]^T$$

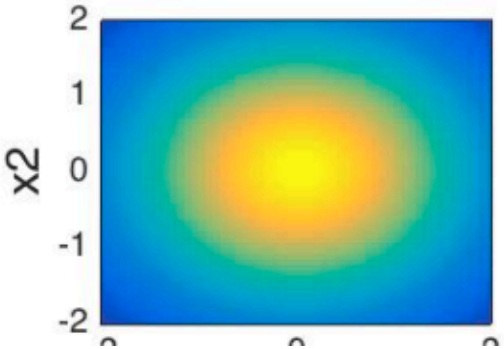




$$\Sigma = \begin{matrix} 1.5 & 0 \\ 0 & 1.5 \end{matrix}$$

$$\mu = [0 \;\; 0]^T$$

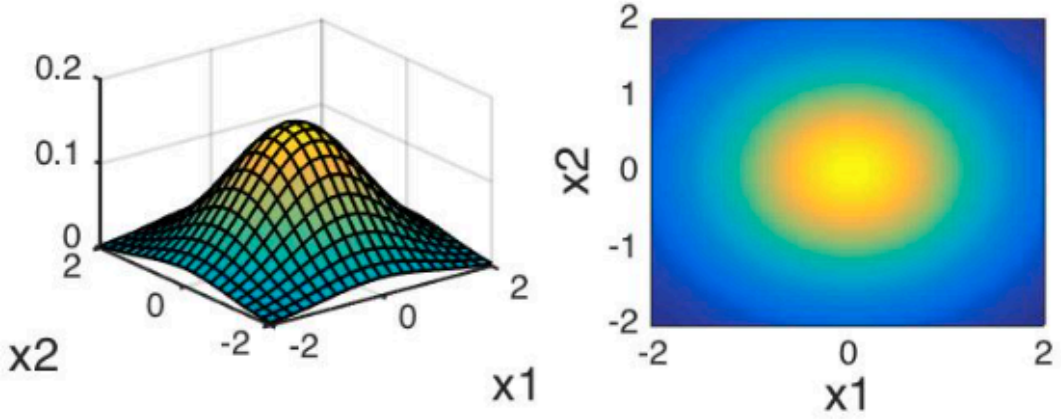




Multivariant Gaussian (Normal examples)

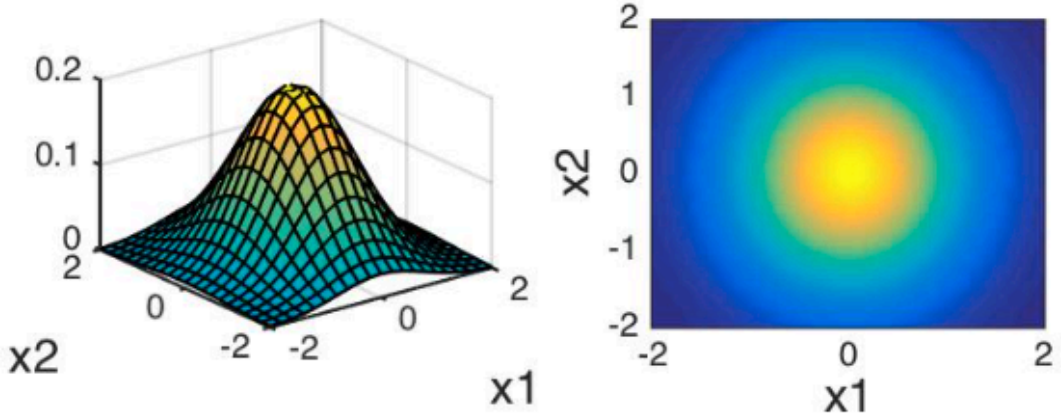
$$\Sigma = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}$$

$$\mu = [0 \;\; 0]^T$$



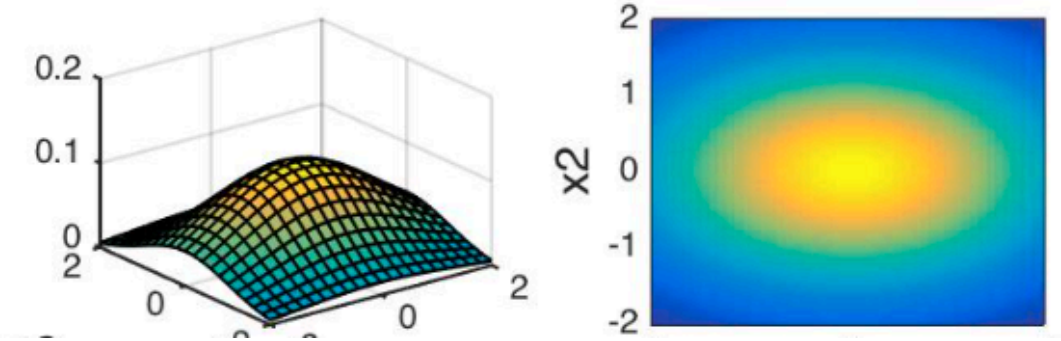
$$\Sigma = \begin{matrix} 0.6 & 0 \\ 0 & 1 \end{matrix}$$

$$\mu = [0 \;\; 0]^T$$



$$\Sigma = \begin{matrix} 2 & 0 \\ 0 & 1 \end{matrix}$$

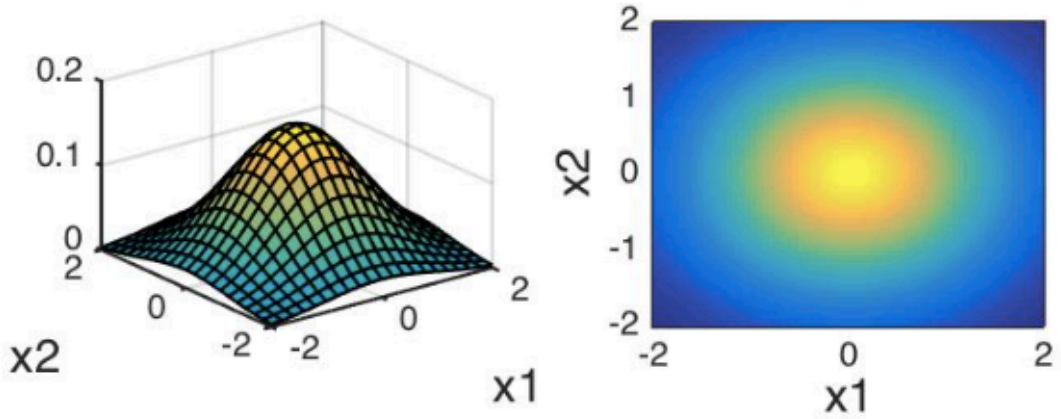
$$\mu = [0 \;\; 0]^T$$



Multivariant Gaussian (Normal examples)

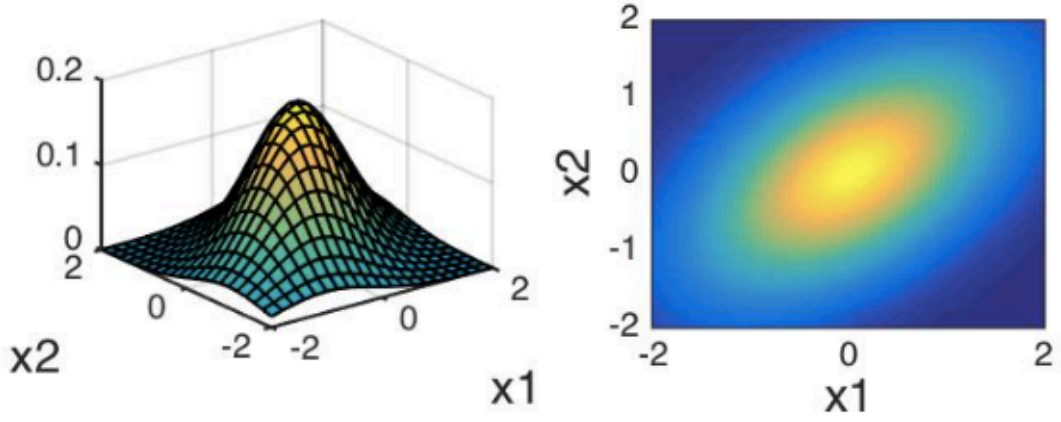
$$\Sigma = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}$$

$$\mu = [0 \;\; 0]^T$$



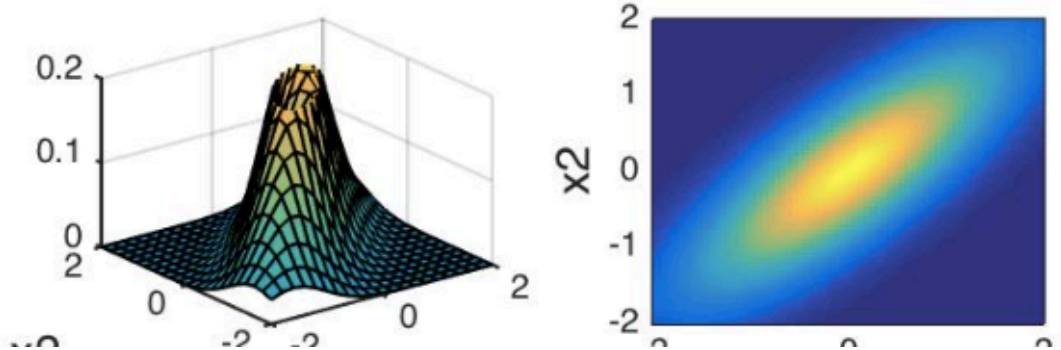
$$\Sigma = \begin{matrix} 1 & 0.5 \\ 0.5 & 1 \end{matrix}$$

$$\mu = [0 \;\; 0]^T$$



$$\Sigma = \begin{matrix} 1 & 0.8 \\ 0.8 & 1 \end{matrix}$$

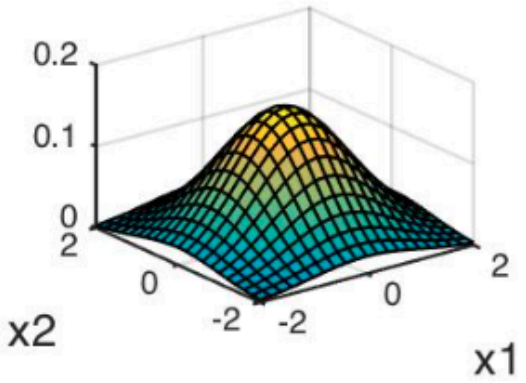
$$\mu = [0 \;\; 0]^T$$

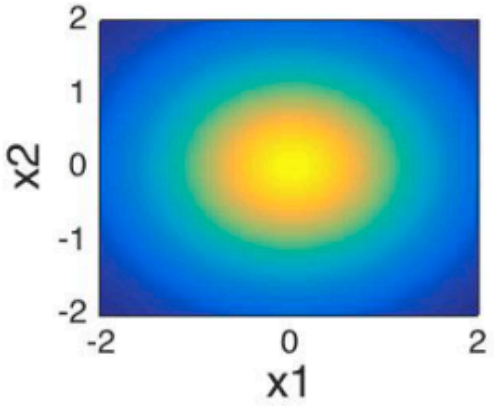


Multivariant Gaussian (Normal examples)

$$\Sigma = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}$$

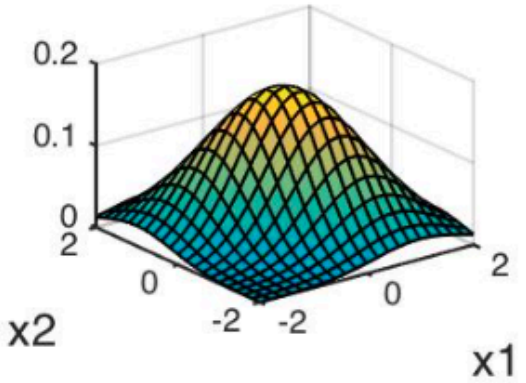
$$\mu = [0 \;\; 0]^T$$

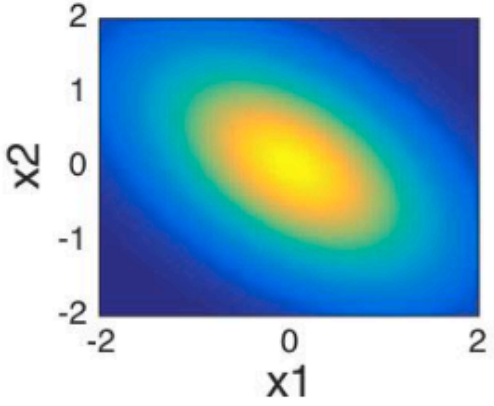




$$\Sigma = \begin{matrix} 1 & -0.5 \\ -0.5 & 1 \end{matrix}$$

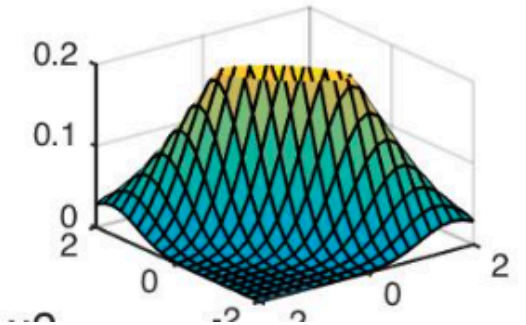
$$\mu = [0 \;\; 0]^T$$

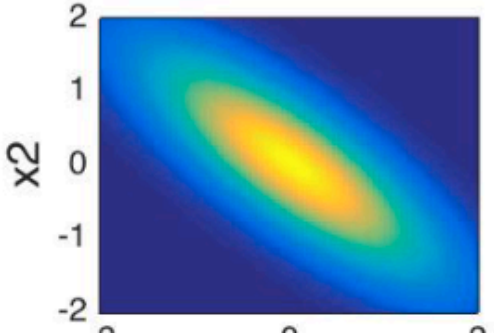




$$\Sigma = \begin{matrix} 1 & -0.8 \\ -0.8 & 1 \end{matrix}$$

$$\mu = [0 \;\; 0]^T$$

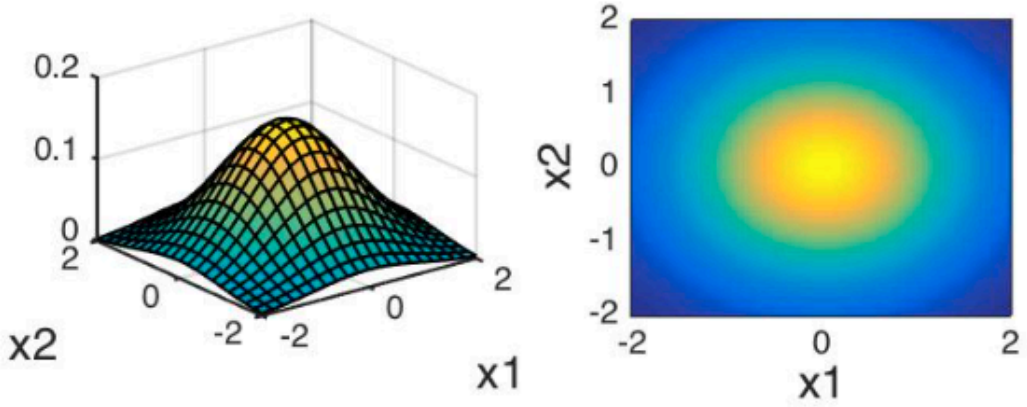




Multivariant Gaussian (Normal examples)

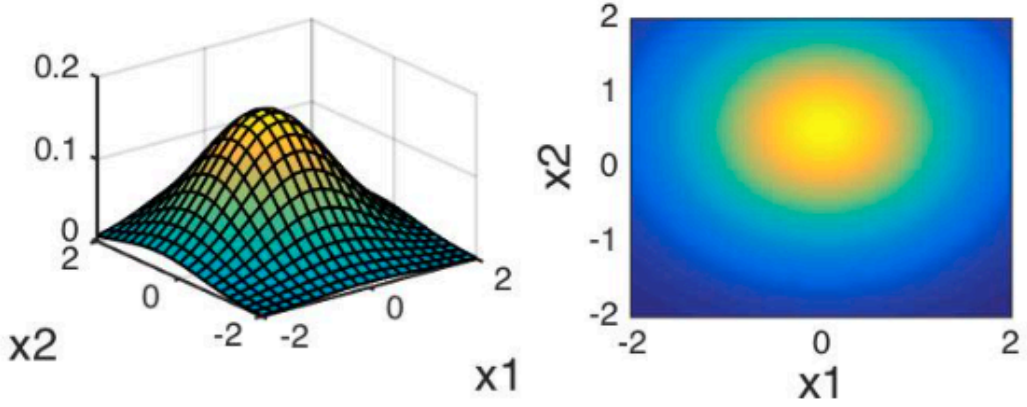
$$\Sigma = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}$$

$$\mu = [0 \;\; 0]^T$$



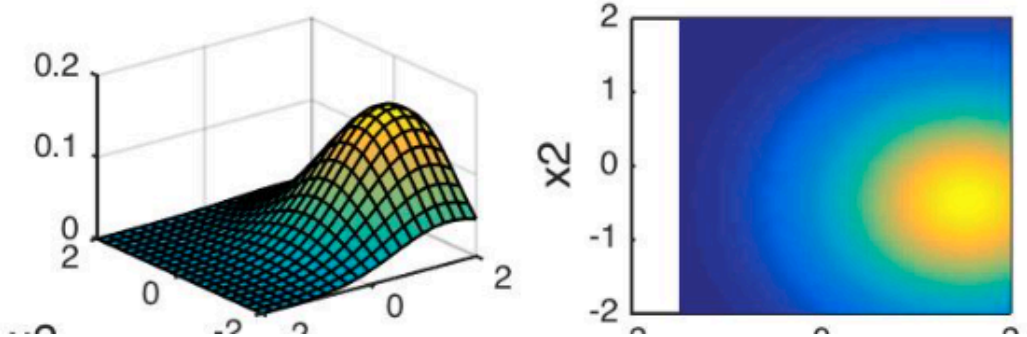
$$\Sigma = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}$$

$$\mu = [0 \qquad 0.5]^T$$



$$\Sigma = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}$$

$$\mu = [1.5 \qquad -0.5]^T$$



Variance

Variance is non-linear but the following holds:

$$\mathrm{Var}(aX) = E(aX - E(aX))^2 = E(aX - aEX)^2 = a^2 E(X - EX)^2 = a^2 \mathrm{Var}(X)$$

$$\mathrm{Var}(X+c) = E(X+c-E(X+c))^2 = E(X-EX+c-c)^2 = E(X-EX)^2 = \mathrm{Var}(X)$$

$$\begin{aligned} \mathrm{Var}(X+Y) &= E(X - EX + Y - EY)^2 \\ &= \underbrace{E(X-EX)^2}_{\mathrm{Var}(X)} + \underbrace{E(Y-EY)^2}_{\mathrm{Var}(Y)} + 2\underbrace{E(X-EX)(Y-EY)}_{\mathrm{Cov}(X,Y)} \end{aligned}$$

So when X, Y are independent we have:

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

(why?)

Law of Large Numbers

Say we have $X_1 \ldots X_n$ i.i.d., where $EX_i = \mu$ and $\text{Var}(X_i) = \sigma^2$.

We want to know the expectation and variance of $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

$$E(\bar{X}_n) = E[\frac{1}{n}\sum_{i=1}^{n} X_i] = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \frac{1}{n} n\mu = \mu$$

$$\text{Var}(\bar{X}_n) = \text{Var}(\frac{1}{n}\sum_{i=1}^{n} X_i) = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}$$

Distribution appears to be "contracting:" as n increases, variance is going to 0.

Law of Large Numbers (LLN)

Recall our variable $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.
We may wonder about its behavior as $n \to \infty$.

The **weak law of large numbers**:

$$\lim_{n\to\infty} P(|\bar{X}_n - \mu| < \epsilon) = 1$$

In English: choose ϵ and a probability that $|\bar{X}_n - \mu| < \epsilon$, I can find you an n so your probability is achieved.

The **strong law of large numbers**:

$$P(\lim_{n\to\infty} \bar{X}_n = \mu) = 1$$

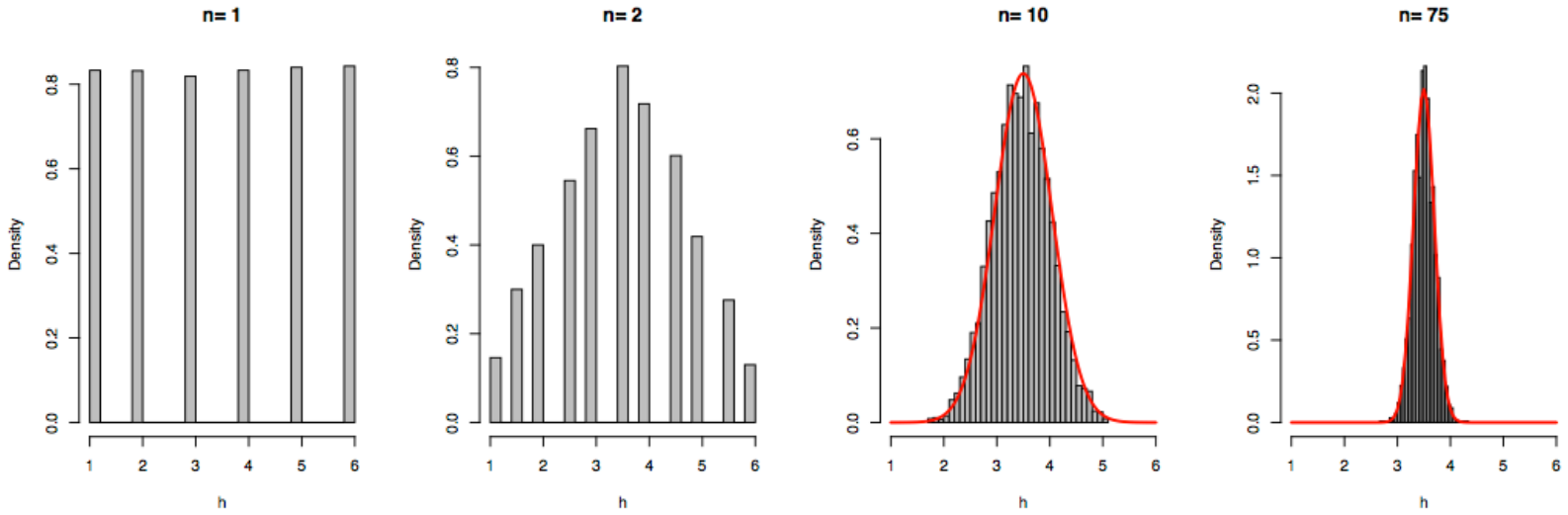
In English: the mean converges to the expectation "almost surely" as n increases.

Central Limit Theorem (CLT)

The distribution of $\bar{X}_n$ also converges weakly to a Gaussian,

$$\lim_{n\to\infty} F_{\bar{X}_n}(x) = \Phi\left(\frac{x-\mu}{\sqrt{n}\sigma}\right)$$

Simulated n dice rolls and took average, 5000 times:



Two kinds of convergence went into this picture (why 5000?):

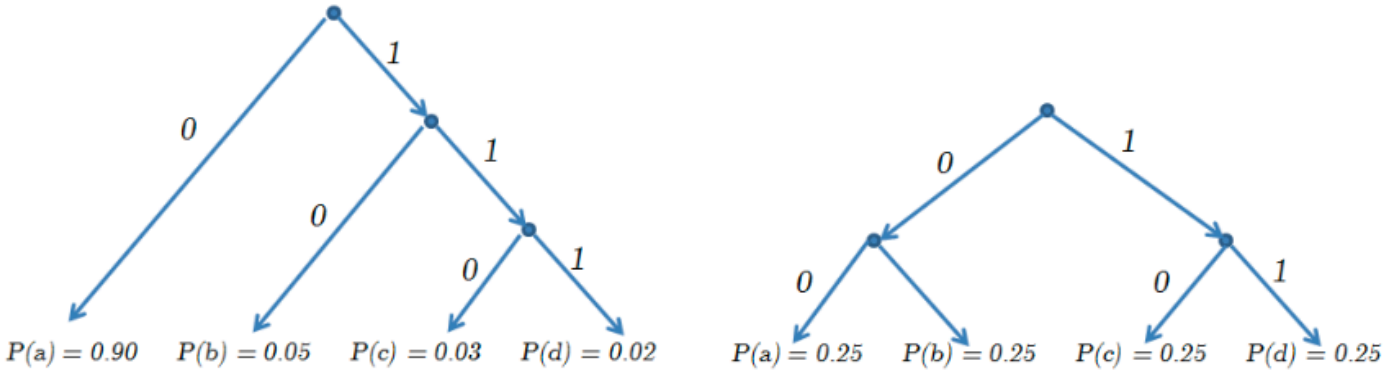
1. True distribution converges to a Gaussian (CLT).
2. Empirical distribution converges to true distribution (Glivenko-Cantelli).

Entropy of a Distribution

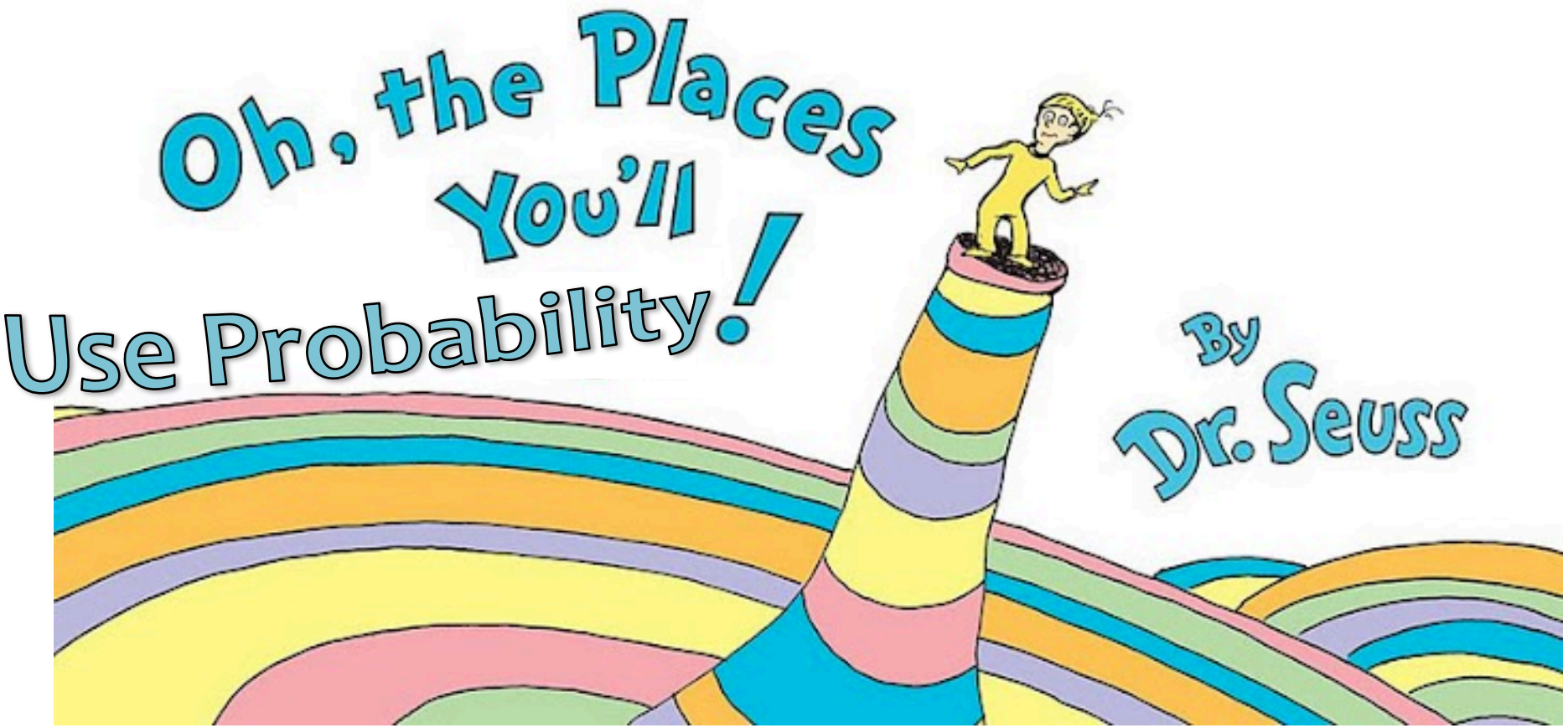
Entropy is a measure of uniformity in a distribution.

$$H(X) = -\sum_x f_X(x) \log_2 f_X(x)$$

Imagine you had to transmit a sample from f_X, so you construct the optimal encoding scheme:



Entropy gives the mean depth in the tree (= mean number of bits).



Use Probability!

Oh, the Places You'll Use Probability!

Supervised Classification

- Naïve Bayes

$$p(y|x_1, x_2, \ldots, x_n) = \frac{1}{Z} p(y) \prod_{i=1}^{n} p(x_i|y)$$

- Logistic regression

$$P(Y = y|X = x; \boldsymbol{\theta}) = p(y|x; \boldsymbol{\theta})$$
$$= \frac{\exp(\boldsymbol{\theta}_y \cdot \mathbf{f}(x))}{\sum_{y'} \exp(\boldsymbol{\theta}_{y'} \cdot \mathbf{f}(x)}$$

Note: This is just motivation – we'll cover these topics later!

Oh, the Places You'll Use Probability!

ML Theory

(Example: Sample Complexity)

- Goal: h has small error over D.

  True error: $err_D(h) = \Pr_{x \sim D}(h(x) \neq c^*(x))$

  How often $h(x) \neq c^*(x)$ over future instances drawn at random from D

- But, can only measure:

  Training error: $err_S(h) = \frac{1}{m}\sum_i I\big(h(x_i) \neq c^*(x_i)\big)$

  How often $h(x) \neq c^*(x)$ over training instances

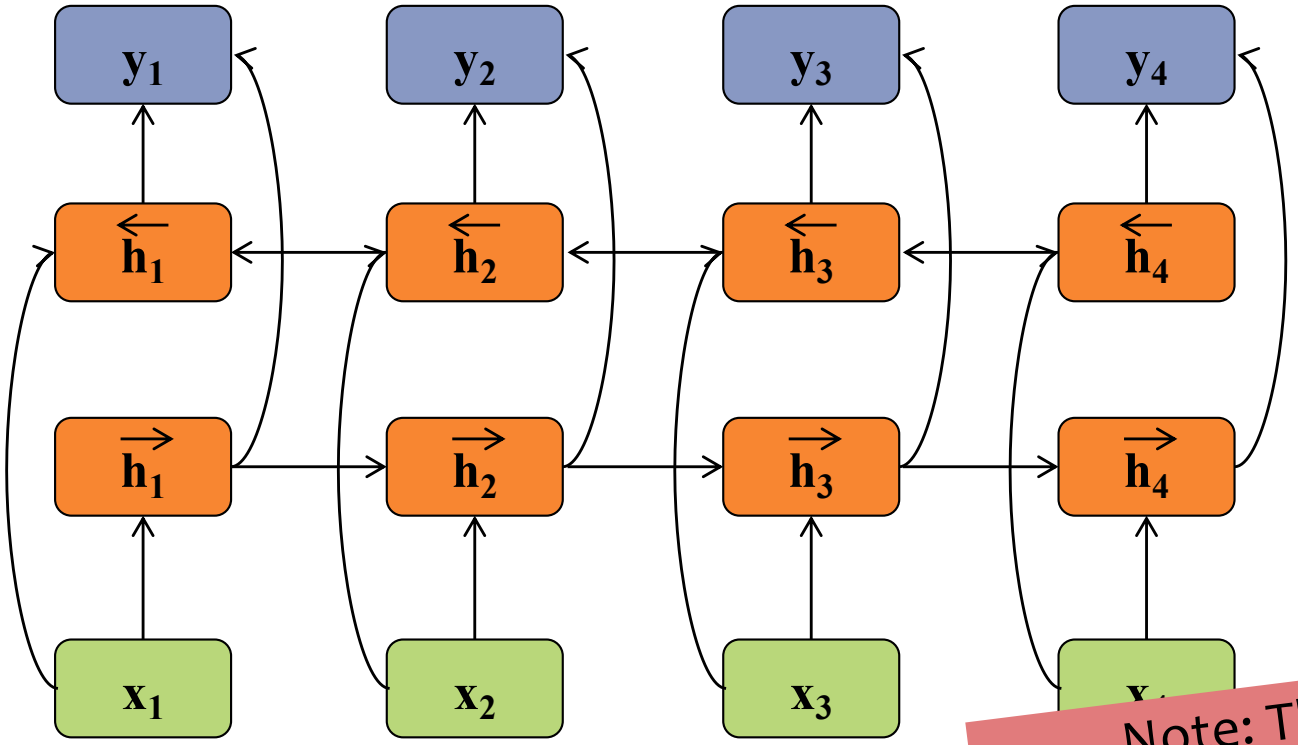
Sample complexity: bound $err_D(h)$ in terms of $err_S(h)$

Note: This is just motivation – we'll cover these topics later!

Oh, the Places You'll Use Probability!

Deep Learning

(Example: Deep Bi-directional RNN)

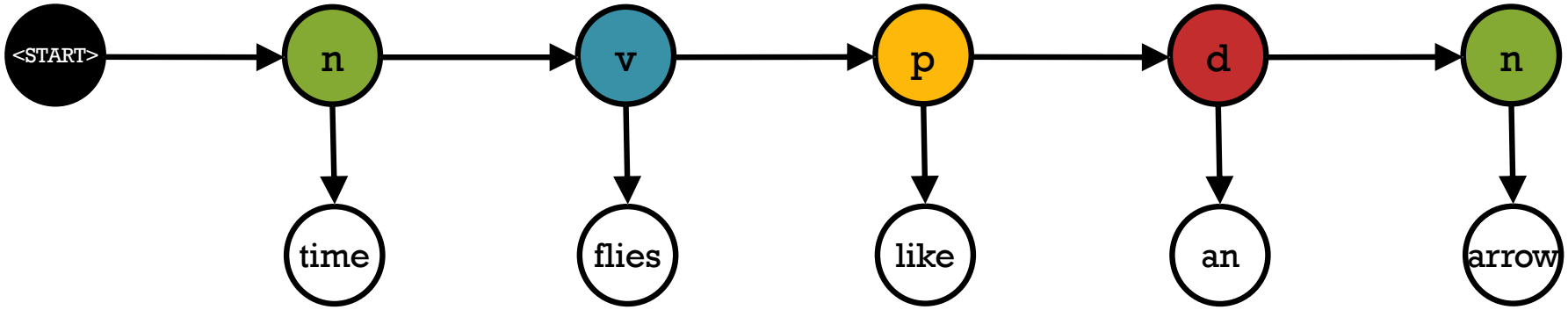


Note: This is just motivation – we'll cover these topics later!

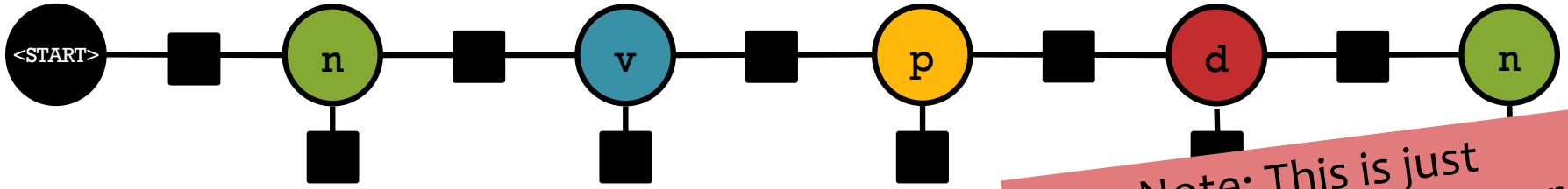
Oh, the Places You'll Use Probability!

Graphical Models

- Hidden Markov Model (HMM)



- Conditional Random Field (CRF)



Note: This is just motivation – we'll cover these topics later!

Summary

1. Probability theory is rooted in (simple) axioms
2. Random variables provide an important tool for modeling the world
3. Our favorite probability distributions are just functions! (usually with interesting properties)
4. Probability and Statistics are essential to Machine Learning